

Solved Problems (IP Addressing)

Problem#1: Find the subnet mask of the network in which the IP address 192.35.128.0/19 is used?

Answer: There are 32 bits in an IP address. The given prefix length 19. This means that there are 19 network bits (in short net bits) and $32-19=13$ host bits.

To find subnet mask of the network, write 19 ones and 13 zeros.

11111111.11111111.11100000.00000000 = 255.255.224.0

The subnet mask is 255.255.224.0

Problem#2: Find the prefix length of an IP address that is used in a network whose subnet mask is 255.252.0.0.

Answer: Convert the given subnet mask into binary:

11111111.11111100.00000000.00000000

There are 14 ones. Thus, there are 14 net bits and $32-14=18$ host bits.

Hence, the prefix length = number of net bits = 14.

Problem# 3: How many host IP addresses are there in the network in which the 192.35.128.0/25 IP address is being used? How many host devices can be configured in this network?

Answer:

Part 1:

The given prefix length is 25. There are 25 net bits and $32-25=7$ host bits.

The number of host IP addresses of the network = $2^{\text{No. of host bits}} = 2^7 = 128$.

There are 128 host IP addresses in the network.

Part 2:

Out of this 128 host IP addresses, the first and last IP addresses are the network IP addresses and Broadcast IP addresses, respectively. Thus, the rest of the $128-2=126$ IP addresses are called usable host IP addresses or valid host IP addresses. Using these 126 usable host IP addresses, 126 devices can be configured.

Problem#4: Find the network and broadcast IP addresses of the network in which the 192.230.5.239/25 IP address is used.

Answer: The prefix length is 25. Hence, there are 25 net bits and $32-25=7$ host bits.

The binary representation of the given IP address is

11000000.11100110.00000101.11101111

Of these 32 bits, the first 25 bits (red) are net bits. The rest of the bits (green) are host bits.

To find the network IP address of the network, set the host bits to 0s (zeros).

11000000.11100110.00000101.10000000 = 192.230.5.128

The network IP address: 192.230.5.128

To find the broadcast IP address of the network, set the host bits to 1s (ones).

11000000.11100110.00000101.11111111 = 192.230.5.255

The broadcast IP address: 192.230.5.255

Problem#5 The third and second last usable host IP addresses are 5.4.160.3 and 5.4.175.253, respectively. Find the subnet mask of the network. Also, find the number of usable host IP addresses of the network.

Answer: If 5.4.160.3 is the third usable host IP address of the network,

Second usable host IP address: 5.4.160.2

First usable host IP address: 5.4.160.1

First IP address of the network (which is called the network IP address): 5.4.160.0

If 5.4.175.253 is the second last usable host IP address of the network,

Last usable host IP address: 5.4.175.254

Last host IP address (which is called the broadcast IP address): 5.4.175.255

We have calculated both the network and broadcast IP addresses. Next, convert them into binary:

Network IP: 00000101.00000100.10100000.00000000

Broadcast IP: 00000101.00000100.10101111.11111111

Check how many bits of the network and broadcast IP addresses match from the beginning.

The first 20 bits (red) of the IP addresses are the same. Hence, these bits are net bits.

In other words, the prefix length is 20.

Then, host bits are $32 - 20 = 12$ bits (black)

Hence, the subnet mask is 11111111.11111111.11110000.00000000=255.255.240.0

The number of host IP addresses of the network: $2^{12}=4096$

First and last host IP addresses are reserved IP addresses. Thus, the number of usable/valid host IP addresses is $4096-2=4094$.

In short, the number of usable host IP addresses is $2^{12} - 2 = 4094$.

Problem#6: The network IP address of a network is 192.168.4.0 and there are 510 usable host IP addresses in the network. Find the broadcast IP address of the network.

Answer: There are 510 usable host IP addresses. Hence, total number of IP addresses is $510+2=512$.

$$\text{No. of IP addresses} = 2^{\text{No. of host bits}}$$

$$\text{No. of host bits} = \log_2(512) = 9$$

No. of net bits: $32-9=25$

Convert the network bits into binary

11000000.10101000.00000100.00000000

Last 9 bits are host bits. To get the broadcast IP address, make these 9 bits 1.

11000000.10101000.00000101.11111111 = 192.168.5.255

The broadcast IP address is 192.168.5.255

Another method:

192.168.4.0 is the first IP address.

Next few IP addresses are 192.168.4.1, 192.168.4.2, 192.168.4.3, 192.168.4.4

192.168.4.253, 192.168.4.254, 192.168.4.255.

From 192.168.4.0 to 192.168.4.255, there are a total of 256 IP addresses.

After 192.168.4.255, we cannot increase in the fourth octet. Then, increment the third octet and make the fourth octet 0. (just like 17, 18, 19, then, 20, 21, 22, 23) Hence, the next IP address will be 192.168.5.0.

Each value of the third octet can produce 256 IP addresses by changing in the fourth octet from 0 to 255 (just like for each value in the tenth position creates 10 values by changing the unit value from 0 to 9. 20, 21, 22, 23, 24, 25, 26, 27, 28, 29)

Thus, 192.168.4.0 to 192.168.4.255 = 256 IP addresses.

192.168.5.0 to 192.168.5.255 = 256 IP addresses.

Since there are 512 IP addresses and $512=256+256$, the last IP would be 192.168.5.255.

Problem#6: There are two subnets, A and B, in a network with the network IP addresses 2.2.2.0/24 (A) and 3.3.0.0/16 (B). Assume that there is one and no multicast group in the network A and B, respectively. Determine the network name and the number of users who will receive the packet if a user of the network A sends a packet to the following IP addresses.

- A) 127.0.0.5
- B) 2.2.2.253
- C) 255.255.255.255
- D) 3.3.3.253
- E) 3.3.255.255
- F) 2.2.2.255
- G) 225.4.3.1

Answer:

- A) This is a loopback address. If a packet is sent to a loopback address, it does not leave the host device. Hence, no device will receive it.
- B) This is a unicast address of the network A. Hence, only one device of this network with this IP address will receive it.
- C) This is the limited broadcast IP address. Since the packet is originated from a device of the network A, all devices of this network will receive it.
- D) This is a unicast address of the network B. Hence, only a device with this IP address of the network B will receive this packet.
- E) This is the last IP address of the network B. Hence, it is called direct broadcast. This packet will be received all devices of the network B.
- F) This is a direct broadcast IP address (as this is the last IP address) of the network. In modern networks, the default behavior on most OSes/routers is to assume this direct broadcast address as the limited broadcast address when the packet originates from the same network. Hence, 2.2.2.255 will be converted into 255.255.255.255 and the packet will be sent to all devices of network A.
- G) This is a multicast address. According to the question, the network has a multicast group, while network B does not. Hence, it can be assumed that 225.4.3.1 is the multicast address of the multicast group of network A. Thus, all the devices of this multicast group will receive this packet.

Problem#7: Write the following IPv6 address in its shortest possible form:

a) 2041:0A00:0000:0000:0000:0000:0000:0000

b) FA80:0000:0000:0000:1234:0000:0000:0078

Answer:

- a) In the second hextet, replace 0A00 with A00 as the leading zeros can be removed.
There sequence of six all-zero hextet (0000), replace all of them by a single ::
Thus, shortest form is 2041:A00::
- b) There are two sequences of all-zero hextet. The first sequence has 3 all-zero hextet while the second sequence (after 1234) has two all-zero hextet. Replace the largest all-zero hextet with a pair of double colons ::. On the other hand, replace each non-zero hextet in the smaller sequence with a single 0 (zero).
Thus, the shortest form: FA80 :: 1234: 0: 0: 78

Problem#8: Expand the following compressed IPv6 addresses to their full hexadecimal form:

a) 2001:0:0:1: :D5

b)::2

Answer:

- a) Replace each single 0 with a all-zero hextet (0000). Additionally, replace 1 and 5 with 0001 and 00D5 as each hextet must consist of four HEX digits.
2001:0000:0000:0001 :: 00D5
Now there are five hextet. There are three missing hextet. Hence, replace, :: with a sequence of three all-zero hextet
2001:0000:0000:0001:0000:0000:0000:00D5
- b) Replace 2 with 0002 and :: with 7 all-zero hextets:
0000:0000:0000:0000:0000:0000:0000:0002